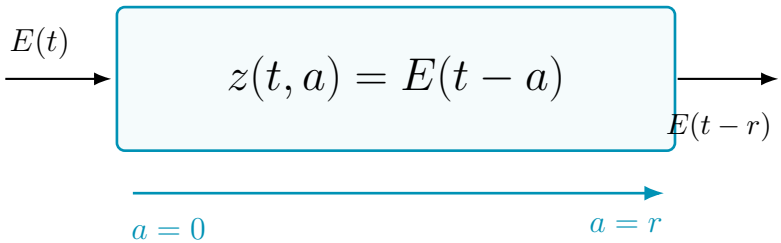
Store the recent activity history



Memory age a, not tissue position.

Refractory memory retained.
Synaptic filtering retained.
No temporal coarse graining.

Transport PDE and refractory fraction

$$\partial_t z + \partial_a z = 0, \qquad a \in [0, r],$$

$$z(t, 0) = E(t),$$

$$R(t) = \int_0^r z(t, a)\, da.$$

Coupled ODEs

$$\dot{U} = -\alpha U + c_1 E + P,$$

$$\tau \dot{E} = -E + (1 - R) S_e(U).$$